

JHU Blockchains and Cryptocurrencies – Final Project

The project's primary goal is to improve the state of the art in some parts of the blockchain and cryptocurrency space, and we encourage groups to follow this path. We will also accept projects where the primary contribution is a **systematization of existing knowledge (SoK)**, but projects of this sort will need to reach a higher bar for evaluation. A simple synthesis, just a recreation of other survey papers, a few copy-pasted paragraphs from a collection of papers, etc., will not be accepted. Survey projects should still contribute some new insight.

The end goal is either a **publishable research paper** or a **solid blockchain application implementation** with an accompanying writeup. All deliverables are submitted on **Gradescope**.

Checkpoints & Deliverables

Checkpoint 0 — Feb 23

Focus: Topic selection & proposal drafting

- ☐ Topic chosen with a clear angle (novel contribution or gap in existing work)
 - ☐ 5-10 relevant papers/resources identified and skimmed
 - ☐ 1-paragraph problem statement written (what's broken, missing, or unexplored?)
 - ☐ If implementation: target system scoped (what it does, what stack)
 - ☐ Proposal draft started (~half page)
 - ☐ Sign up your team: see Piazza for sign-up sheet
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Checkpoint 1 — March 2 — Deliverable: Project Proposal

Focus: Proposal submission & deep research begins

Submit to Gradescope: Project Proposal

Your project proposal must contain the following elements:

1. **Team members (1-3 people per team)**
2. **A detailed description of what you are investigating**
3. **Discussion on why your project is novel** (include references to prior work)

The first item shouldn't be too hard. You get to pick your group and we want to know who you picked. This project is a large effort over the entire semester.

The second and third items are a bit trickier. We want to know what you're going to investigate in your project. This project is going to require you to synthesize and create something new. This means that your proposal needs to reference prior work. For example:

- Are you building off of existing work? If so, how are you proposing to extend it? Did something fall short in prior work?
- Are you addressing a shortcoming of prior work? If so, what is the shortcoming? How are you proposing to remedy it?
- Is there anything you need to understand or research further before you can begin?

A good proposal will be about **a half page** (not including your bibliography) in 12pt font, single-spaced. Your proposal should not be more than a full page.

If you do your proposal right, you've just written most of your prior work section for your paper.

Criteria:

- ☐ Proposal submitted on Gradescope
 - ☐ 5-10 papers read (not just skimmed), key findings noted
 - ☐ Related work section drafted (who did what, what's the gap)
 - ☐ If research: concrete research question(s) or hypothesis defined
 - ☐ If implementation: architecture designed, repo scaffolded, core dependencies chosen
 - ☐ Paper outline with section headers and bullet points for each section
 - ☐ If implementation: basic prototype running (proof of concept)
 - ☐ Sign up for Checkpoint 1 meeting: see Piazza for sign-up sheet
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Checkpoint 2 — March 23 — Deliverable: Draft 1

Focus: Substantive progress, show some results

Submit to Gradescope: Draft 1

The primary objective of this draft is proof that you are making substantive progress. Substantive progress includes some experimental results, proof sketches, etc. Each project is unique and will be different, but we suggest the following as good baselines:

- **Research project:** 5-7 pages (not including bibliography), 10-15 sources
- **SoK project:** 10-14 pages (not including bibliography), 20-30 sources

Drafts must be submitted in **USENIX format**. Templates are available from USENIX and Overleaf.

Criteria:

- ☐ Draft 1 submitted on Gradescope (5-7 pages research / 10-14 pages SoK)
 - ☐ 10-15 sources cited
 - ☐ If research: preliminary results, data, or proof sketches included
 - ☐ If implementation: core functionality working, architecture section written
 - ☐ Introduction, related work, and methodology sections solid
 - ☐ Figures/tables/diagrams started (system architecture, data flow, results)
 - ☐ Clear articulation of contribution: "we show/build/prove X"
 - ☐ Sign up for Checkpoint 2 meeting: see Piazza for sign-up sheet
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Checkpoint 3 — April 6 — Deliverable: Draft 2

Focus: Polished draft ready for peer review

Submit to Gradescope: Draft 2 (including anonymous version for peer review)

Draft 2 should be a **polished rough draft**. You should be comfortable with others reading it and providing feedback, because you will be submitting this for a peer review. The paper should be mostly done.

- **Research project:** 10-13 pages (not including bibliography), 20-30 sources
- **SoK project:** 20-30 pages (not including bibliography), 40-60 sources

Drafts must be submitted in **USENIX format**.

Criteria:

- ☐ Draft 2 submitted on Gradescope (10-13 pages research / 20-30 pages SoK)
 - ☐ 20-30 sources cited
 - ☐ Anonymous version submitted on Gradescope for peer review
 - ☐ If research: complete results section with analysis/evaluation
 - ☐ If implementation: full system working, benchmarks or evaluation included
 - ☐ All major sections complete (intro, related work, methodology, results, discussion)
 - ☐ Figures/tables polished and referenced properly
 - ☐ Paper tells a coherent story from problem to solution to evaluation
 - ☐ Sign up for Checkpoint 3 meeting: see Piazza for sign-up sheet
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Checkpoint 4 — April 13-20 — Deliverable: Peer Review

Focus: Peer review submission, incorporate feedback & prepare presentation

Submit to Gradescope (April 13): Peer Review

Each person in the class will be randomly assigned 1-2 projects to read and review according to a set template (template forthcoming). You will be graded on your thoroughness and quality of feedback. **This is an individual assignment.**

Presentations (April 14 - April 27): Your group will give a **5-7 minute presentation** in class. Sign ups for time slots are forthcoming.

Criteria:

- ☐ Peer review submitted on Gradescope (April 13)
 - ☐ Peer review feedback on your own paper read and addressed
 - ☐ Weak sections identified and rewritten
 - ☐ If implementation: code cleaned up, README written, reproducible
 - ☐ 5-7 minute presentation prepared (slides done)
 - ☐ Presentation rehearsed at least once
 - ☐ Key talking points: problem, approach, results, takeaway
 - ☐ Near-final draft in progress
 - ☐ Sign up for Checkpoint 4 meeting: see Piazza for sign-up sheet
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Checkpoint 5 — April 27 — Deliverable: Final Draft

Focus: Ship it

Submit to Gradescope: Final Draft + code/artifacts

Your final draft is your completed and polished project. We intend for this to take the place of your final exam. The ideal final project is a paper that is publishable at an academic blockchain or cryptography conference.

Code or other artifacts associated with your project should also be submitted. Code should be in a state where TAs can easily run/reproduce results by following instructions in a README.

Criteria:

- ☐ Final draft submitted on Gradescope, fully polished
 - ☐ All peer review and TA/prof feedback incorporated
 - ☐ If implementation: code + README submitted, TAs can reproduce results
 - ☐ Paper proofread for grammar, clarity, formatting (USENIX template)
 - ☐ Bibliography complete and properly formatted
 - ☐ If publishable: identified 1-2 target venues (workshop/conference)
 - ☐ Honest self-assessment: is this publishable, or what would it take to get there?
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Ideas

There are two main ways to get ideas for your project. The first is to look at real blockchain and cryptocurrency papers (highly recommended for students in 601.641). The second is to peruse our list of suggested topics.

Conferences

- USENIX Security
- IEEE S&P
- ACM CCS
- FC
- IACR Conferences
- IACR ePrint
- ArXiv
- Oakland SoK

Some tips for finding papers:

- Command-F (or Ctrl-F) and search for keywords like "blockchain" and "cryptocurrency"
- Pay special attention to tracks like *Blockchain I* and *Blockchain II*
- Other years are also good to look at

Suggested Topics

For each of the following topics, if the scope is not defined precisely you should read the papers and systematize the progress in the topic. Any attempts to improve existing work are encouraged and would be greatly valued in general and for grading. We are also attaching resources to help you select topics. The blog posts are a nice first introduction, and the papers are what you should read and aim to improve upon for the project. The list of papers is not exhaustive by any means, you should look at and search for papers online. Feel free to discuss this with us.

Payment Channels

Detailed analysis of shortcomings/vulnerabilities of existing payment channels like Lightning network. Survey of recent proposals that have strong cryptographic guarantees, their potential limitations, and new directions.

- https://en.bitcoin.it/wiki/Payment_channels
 - <https://academy.binance.com/en/articles/blockchain-scalability-sidechains-and-payment-channels>
 - <https://arxiv.org/abs/2002.06564>
 - <https://eprint.iacr.org/2017/820.pdf>
 - <https://eprint.iacr.org/2016/701>
 - <https://eprint.iacr.org/2020/1613.pdf>
 - <https://www.ndss-symposium.org/ndss-paper/anonymous-multi-hop-locks-for-blockchain-scalability-and>
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Mixers and Privacy Attacks

Mixers are an early attempt at adding anonymity on top of blockchains. However, recent work has broken older mixers and allowed for the deanonymization of their users. Look at the shortcomings of such mixers, attacks on them, and whether modern mixers are more resilient. Advanced network analysis techniques have also been used to systematically deanonymize addresses. Hence, privacy attacks on popular blockchains are also in scope.

- <https://eprint.iacr.org/2021/1088>
 - <https://petsymposium.org/popets/2022/popets-2022-0091.php>
 - <https://arxiv.org/abs/2005.03535>
 - <https://arxiv.org/abs/2007.00764>
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ZK Rollups

Improving blockchain scalability through zk rollups: ZK-Rollups are one of the options being developed for layer 2 construction that increases scalability through mass transfer processing rolled into a single transaction.

- <https://docs.ethhub.io/ethereum-roadmap/layer-2-scaling/zk-rollups/>
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Verifiable Random Functions and Randomness Beacons

Verifiable random functions, verifiable delay functions, randomness beacons and their applications: randomness is an essential requirement in blockchains for multiple purposes. Any attacks on randomness can seriously compromise blockchains. Hence, verifiable randomness is a highly desirable functionality.

- <https://medium.com/algorand/algorand-releases-first-open-source-code-of-verifiable-random-function-1234567890>
 - https://people.csail.mit.edu/silvio/Selected%20Scientific%20Papers/Pseudo%20Randomness/Verifiable_Random_Functions.pdf
 - <https://tools.ietf.org/id/draft-irtf-cfrg-vrf-06.html>
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Stablecoins

An analysis of stablecoins. Possible new results: Methods of analyzing how stablecoin can be achieved, new ways to build algorithmic or collateralized stablecoins. Possible topics to include in a survey: How are stablecoins built? How have they failed? Algorithmic Stablecoins security, vulnerabilities, assumptions, and challenges.

- <https://www.coindesk.com/what-are-stablecoins>
 - <https://vitalik.eth.limo/general/2022/05/25/stable.html>
 - <https://arxiv.org/abs/1910.10098>
 - See Circle, Celo, Tether, etc.
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Scaling and Data Availability

How do we scale blockchains and ensure that critical transaction data remains available? Many proposals have been offered for this. They include "DA" layers and enhanced scaling proposals.

- <https://a16zcrypto.com/posts/article/an-overview-of-danksharding-and-a-proposal-for-improvement-of-eth-1-data-availability>
- <https://ethereum.org/roadmap/danksharding/>
- <https://ethereum.org/developers/docs/scaling/>

Privacy-Preserving Smart Contracts

These proposals of smart contracts provide privacy of transactions (and other actions on the blockchain) from the public view. How do we build them, and how do they fail?

- <https://ieeexplore.ieee.org/document/7546538>
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Secure Bridges

Bridges allow for moving funds from one blockchain to another. They have a fraught history, with many hacks and breakages. Possible new results: security analysis of previously unanalyzed bridges, new methods for constructing bridges. Possible topics to include in a survey: How do we build bridges, and how do they fail? How can we do better?

- <https://eprint.iacr.org/2021/1589>
 - <https://rdi.berkeley.edu/zkp/zkBridge/uploads/paper.pdf>
 - <https://arxiv.org/pdf/2210.16209.pdf>
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MEV (Maximal Extractable Value)

MEV refers to miners ordering transactions in the blocks they mine to extract value for themselves. What does the current MEV landscape look like, what risks or benefits does it produce, and how could we stop it? Possible new results could be better ways to achieve MEV, new ways of analyzing/modeling MEV, and methods for combatting it.

Sharding

Sharding is a method for scaling blockchains. How does sharding work, how sound is it, and what are the barriers to deploying it?

- <https://vitalik.ca/general/2021/04/07/sharding.html>
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CRS Ceremonies

Some blockchains and applications require a CRS (common reference string) to be generated honestly to preserve security. Many groups have gone about this in different ways. How did such ceremonies go? What parameters did they use? How long did they take? Can we do them better?

- <https://ceremony.ethereum.org/>
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Private AMMs

DeFi is filled with AMMs (Automated Market Makers). Recent work has attempted to make such smart contracts private. How do these implementations work, are they private, and can they be improved?

- <https://eprint.iacr.org/2021/743>
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Stellar Consensus

The Stellar blockchain and their consensus protocol.

- <https://stellar.org/>
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ZKVMs

A ZKVM is a virtual machine in which correct execution can be verified using ZK proofs. Dive into how a specific ZKVM implementation (such as Miden) works and suggest ways to improve it.

- <https://github.com/0xPolygonMiden/miden-vm>
 - <https://hackmd.io/@sin7y/HkIELMUu9>
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Polygon ZKEVM

The Polygon ZKEVM main net is close to launching. Dive into how their ZKEVM works and if it is truly production ready.

- <https://polygon.technology/solutions/polygon-zkevm/>
 - <https://wiki.polygon.technology/docs/zkEVM/introduction/>
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On-chain oracles

How do we deliver data to blockchains, so that they can see a picture of the world?

- <https://research.chain.link/whitepaper-v1.pdf>
 - <https://forum.dfinity.org/t/zkTLS-an-alternative-to-native-https-outcalls-for-private-or-big-data/55352>
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IPFS and Proof of Storage

Dive deep into the solutions, how they work, open problems, and how to improve existing solutions.

- <https://docs.ipfs.tech/>
 - [https://golden.com/wiki/Proof-of-storage_\(PoS\)-MN4DJY3](https://golden.com/wiki/Proof-of-storage_(PoS)-MN4DJY3)
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Blockchain TPS Analysis

The theoretical TPS (transactions per second) of a blockchain and the actual TPS are often very different. Researchers have recently done work to analyze this gap. What is the real TPS of blockchains not studied in their work?

- <https://gramoli.github.io/pubs/Eurosys23-Diablo.pdf>
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Crypto-Heavy Topics

The following are crypto-heavy projects, more suitable for someone with relevant background.

Threshold Cryptography in Blockchains

Threshold cryptography allows for actions to be taken based on the decision of a minimum threshold of members in a group. For example: in a cryptocurrency wallet controlled by a group of n people, for some threshold t , if more than $t - 1$ people sign a transaction then the tokens can be spent successfully. Look deeper into threshold cryptography literature (as well as multi-signatures) related to blockchains for recent improvements, attacks and challenges. Systematize existing literature and work on ways to solve some of the existing challenges. One interesting project that can be the project's main focus is NuCypher.

- <https://ieeexplore.ieee.org/document/9152696>
 - <https://eprint.iacr.org/2019/1328>
 - <https://eprint.iacr.org/2019/523.pdf>
 - <https://eprint.iacr.org/2019/514.pdf>
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Vector Commitments and Succinct Proofs of Membership

Vector commitments allow committing to a set of values while hiding them. At the same time proofs of membership can be produced for values in the committed vector. This becomes very useful for cryptocurrencies, for example, you should prove that your token is part of the unspent tokens (UTXO) set. Recent work aims to improve the efficiency of this primitive.

- <https://alinush.github.io/2020/11/24/Catalano-Fiore-Vector-Commitments.html>
 - <https://eprint.iacr.org/2011/495.pdf>
 - <https://eprint.iacr.org/2018/1188.pdf>
 - <https://eprint.iacr.org/2020/149.pdf>
 - <https://eprint.iacr.org/2020/419.pdf>
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ZK-SNARKs Trust Models

What sort of trust models have we used for deploying ZK-SNARKs?